1) Compute the expectation of a fair coin, where

Heads $= 0$ $\quad$ $P(\text{Heads}) = 1/2$

Tails $= 1$ $\quad$ $P(\text{Tails}) = 1/2$

$$X = \{0, 1\} \qquad E[X] = \sum_{i=0}^{2} x_i P(x_i) = \frac{1}{2}(0+1) = 1/2$$

2) A random variable takes values $X = \{1, 3, 5\}$ with equal probability. Compute $E[X]$

$$E[X] = \frac{1}{3}(1+3+5) = 3$$

3) Center the values from exercise 2.

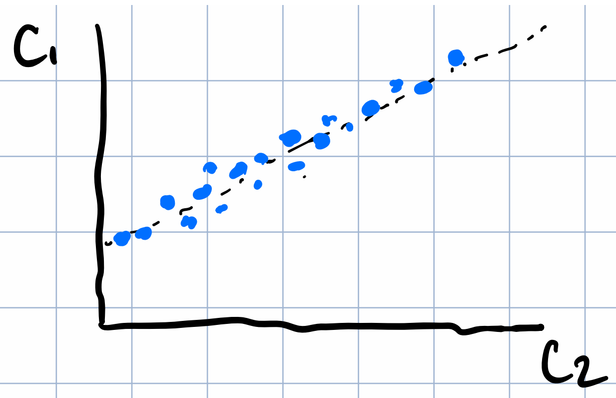
$$\tilde{X} = X - E[X] = \{-2, 0, +2\}$$

4) Compute the variance from exercise 2

$$\text{Var}(X) = E[(X - E[X])^2] = E[\{-2,0,2\}\cdot\{-2,0,2\}] = \frac{8}{3}$$

5) Suppose two variables always increase together. Should their covariance be positive, negative, or zero? Explain.

If two variables always increase together, their covariance will be positive.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